SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Dhivya Dharshini S *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dhivya

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	6
Experimental Design	30	7
Use of Tools	20	5
Analysis of Results	15	3
Documentation	10	2

Total Marks 25 /30

Signature of the Faculty

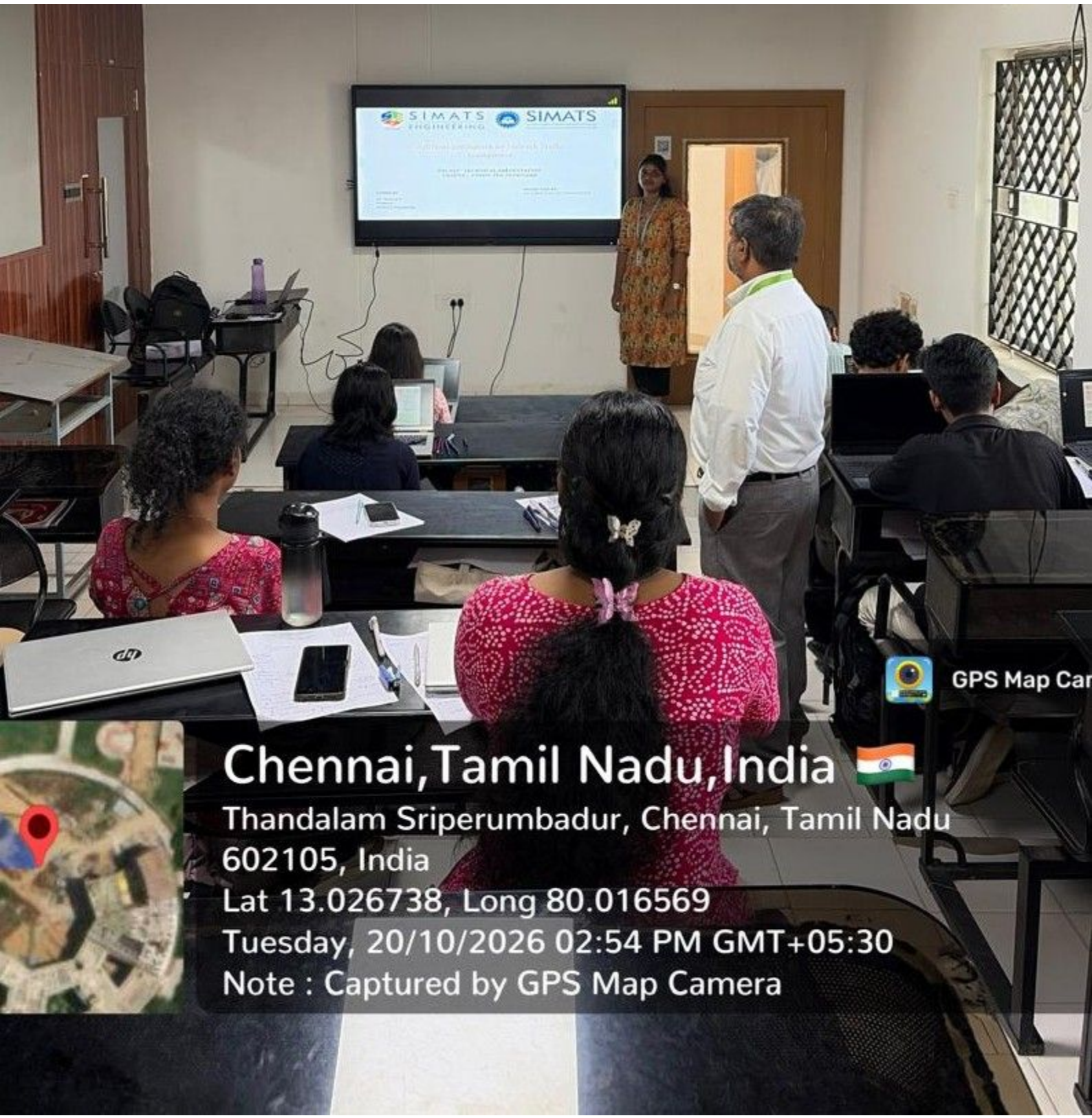
TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation


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GPS Map Cam
Chennai,Tamil Nadu,India
Thandalam Sriperumbadur, Chennai, Tamil Nadu
602105, India
Lat 13.026738, Long 80.016569
Tuesday, 20/10/2026 02:54 PM GMT+05:30
Note : Captured by GPS Map Camera



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Artificial Intelligence For Network Traffic Management

C01 AT3 - TECHNICAL PRESENTATION
CSA0710 - COMPUTER NETWORK

GUIDED BY

Dr. Shankar R,
Professor,
SIMATS Engineering

PRESENTED BY:
192524448 DHIVYA DHARSHINI S

INTRODUCTION

- Modern networks carry massive, fast-changing traffic from cloud, video, IoT and 5G applications.
- Traditional static, rule-based traffic management cannot adapt to sudden congestion or new traffic patterns.
- Artificial Intelligence (AI) and Machine Learning (ML) learn from traffic data to classify, predict and control traffic automatically.
- AI-driven traffic management improves throughput, reduces latency and strengthens network security.

PROBLEM STATEMENT

Growing Load

Exponential growth in users, devices and bandwidth-hungry applications overloads links.

Static Policies

Fixed routing and QoS rules react slowly and cannot handle bursty or encrypted traffic.

Manual Effort

Administrators struggle to detect anomalies and tune the network in real time.

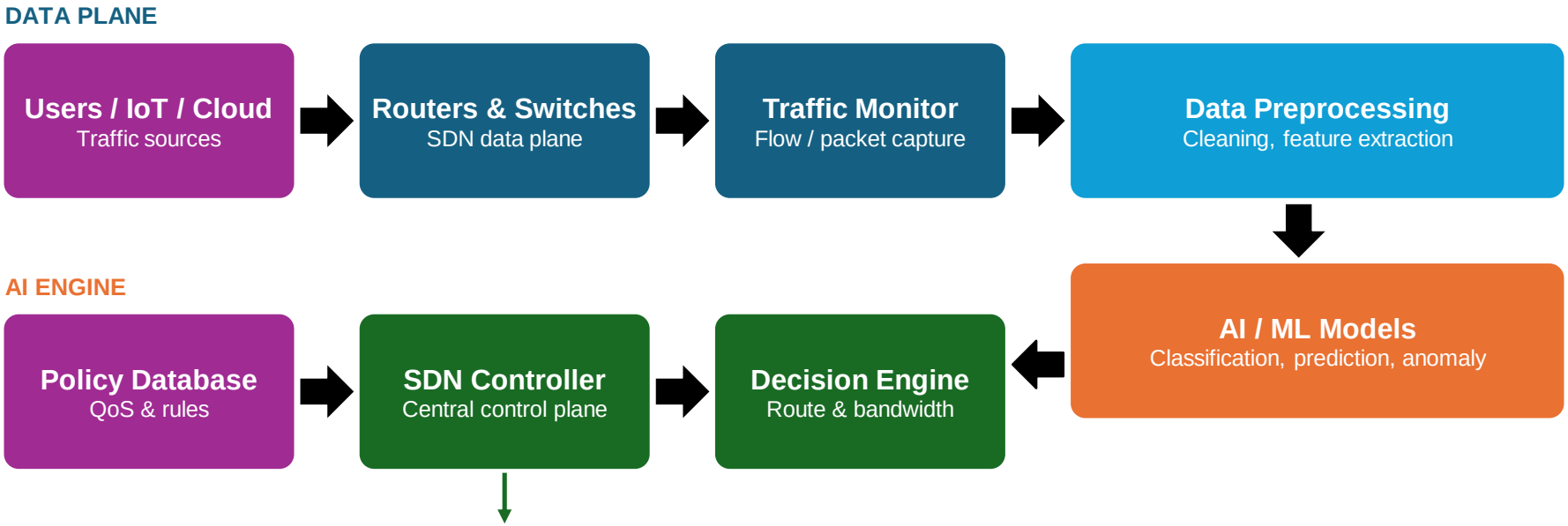
OBJECTIVES

- Classify network traffic accurately using AI/ML models.
- Predict traffic load and congestion before it occurs.
- Optimize routing and bandwidth allocation dynamically.
- Guarantee Quality of Service (QoS) for critical applications.
- Detect anomalies and attacks such as DDoS in real time.
- Reduce manual intervention through automated, self-learning control.

TRADITIONAL VS AI-BASED TRAFFIC MANAGEMENT

Aspect	Traditional	AI-Based
Decision making	Fixed rules & thresholds	Learned from data
Adaptability	Manual reconfiguration	Automatic, real time
Congestion handling	Reactive	Predictive & proactive
Encrypted traffic	Port/payload inspection fails	Flow-feature classification
Security	Signature based	Anomaly & behaviour based
Scalability	Limited	High with SDN integration

ARCHITECTURE DIAGRAM



Optimized routes & QoS rules pushed back to network devices (feedback loop)

MODULES

1. Data Collection

Captures packets, flows and SNMP/telemetry data.

2. Preprocessing

Cleans data and extracts features (size, duration, ports).

3. Traffic Classification

Identifies applications and flow types using ML.

4. Prediction & Anomaly

Forecasts load and detects attacks or faults.

5. Decision & Control

Chooses routes and QoS, enforced by SDN controller.

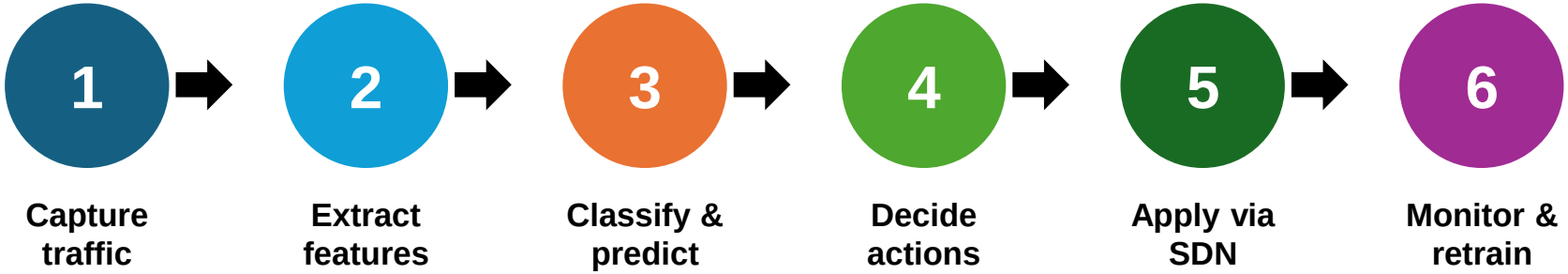
6. Monitoring & Feedback

Measures results and retrains models continuously.

AI TECHNIQUES USED

Technique	Purpose in Traffic Management
Supervised Learning	Traffic classification, intrusion detection
Unsupervised Learning	Anomaly detection, flow clustering
Deep Learning (LSTM, CNN)	Traffic forecasting, encrypted traffic analysis
Reinforcement Learning	Dynamic routing, congestion control, bandwidth allocation
Federated Learning	Privacy-preserving distributed model training

WORKING PROCESS



The cycle repeats continuously, so the network keeps learning and adapting to changing traffic.

ADVANTAGES

Higher Throughput

Better link utilization and load balancing.

Lower Latency

Proactive congestion avoidance.

Guaranteed QoS

Priority for voice, video and critical apps.

Stronger Security

Real-time anomaly and DDoS detection.

Automation

Less manual configuration and fewer errors.

Scalability

Adapts as the network and traffic grow.

APPLICATIONS

- Data centers and cloud networks – load balancing and resource allocation.
- 5G / 6G and telecom networks – network slicing and traffic steering.
- Software-Defined Networking (SDN) and enterprise campus networks.
- Smart cities and Internet of Things (IoT) deployments.
- Network security – intrusion and DDoS detection.
- Content delivery, video streaming and online gaming platforms.

CHALLENGES AND FUTURE SCOPE

Challenges

- Need for large, high-quality labelled datasets
- Real-time processing overhead
- Privacy of traffic data
- Explainability of AI decisions
- Adversarial attacks on models

Future Scope

- Self-driving (intent-based) networks
- Edge AI for faster local decisions
- Federated learning across operators
- Explainable AI for trusted control
- AI-native 6G networks

CONCLUSION

- AI transforms network traffic management from reactive and manual to predictive and automated.
- ML-based classification, prediction and anomaly detection improve performance, QoS and security.
- Integration with SDN enables real-time enforcement of AI decisions.
- With solutions to data, privacy and explainability challenges, AI will be central to future intelligent networks.